Classification Model Accuracy

- classifier RandomForestClassifier has Average CV score is 72.92% with std of 0.02946.
- classifier SVC has Average CV score is 68.75% with std of 0.05103.
- classifier DecisionTreeClassifier has Average CV score is 64.58% with std of 0.02946.
- classifier Lreg has Average CV score is 62.50% with std of 0.135.
- KNN Average CV score: 70.83% ± 0.0295

Regression Model Accuracy

- Random Forest Regressor CV R^2: −0.341 ± 0.355
- SVR CV R^2: −0.257 ± 0.164
- XGBoost Regressor CV R^2: −0.181 ± 0.249

Ranking Models

❖ **LightGBM Ranker**

- LightGBM Mean Rank IC: 0.3557073317848881
- LightGBM Top-3 Hit Rate: 0.5
- LightGBM IC-IR: 1.20567105786513

Label	Name	Average Rank Score	Overall Rank
0	CARG	-0.0568	4
1	CTC	-0.4007	7
2	DIAL	-0.0404	3
3	DIST	0.007	2
4	MELS	-1.5814	10
5	JKH	-1.2669	8
6	LOLC	-1.4557	9
7	COMB	-0.3206	6
8	HNB	0.1992	1
9	SAMP	-0.2271	5

❖ XGBoost Ranker

- XGBoost Mean Rank IC: 0.5880681818181818.
- XGBoost Top-3 Hit Rate: 0.625
- XGBoost IC-IR: 5.751351213452055

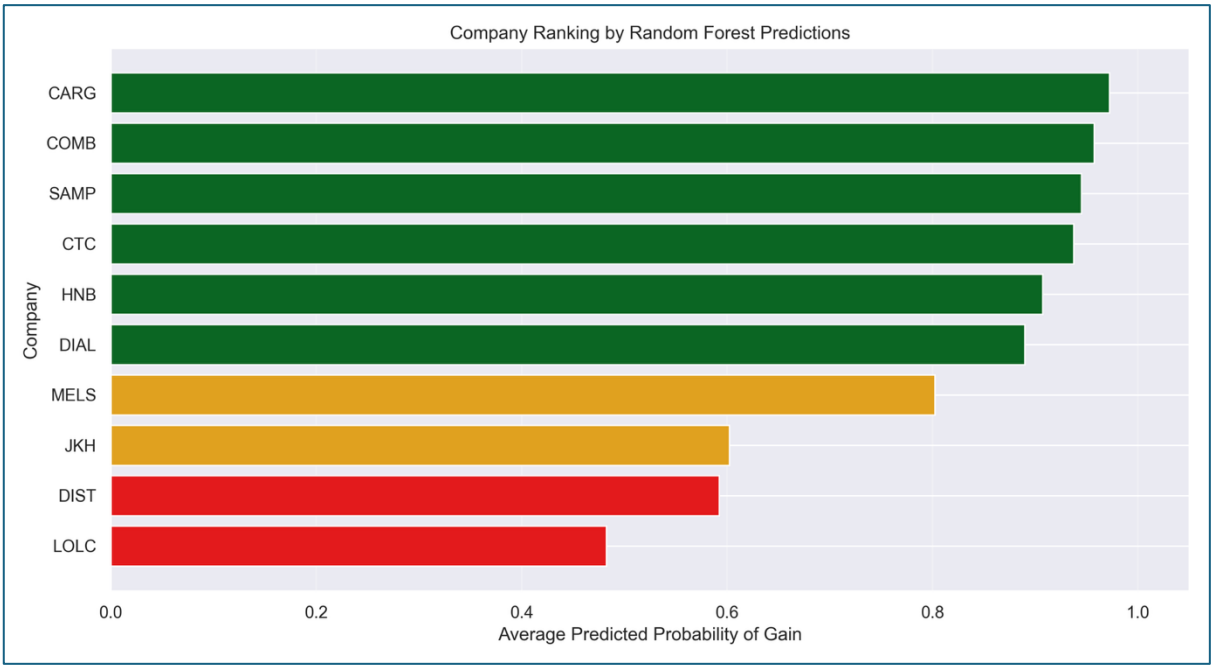
Label	Name	Average Rank Score	Overall Rank
0	CARG	-0.1005	5
1	CTC	-0.0257	4
2	DIAL	0.7255	1
3	DIST	0.1645	3
4	MELS	-0.2407	6
5	JKH	-2.3218	10
6	LOLC	-0.5496	7
7	COMB	-0.6443	9
8	HNB	0.2973	2
9	SAMP	-0.5695	8

Label	Name	LightGBM Score	LightGBM Rank	XGBoost Score	XGBoost Rank
0	CARG	-0.0568	4	-0.1005	5
1	CTC	-0.4007	7	-0.0257	4
2	DIAL	-0.0404	3	0.7255	1
3	DIST	0.0070	2	0.1645	3
4	MELS	-1.5814	10	-0.2407	6
5	JKH	-1.2669	8	-2.3218	10
6	LOLC	-1.4557	9	-0.5496	7
7	COMB	-0.3206	6	-0.6443	9
8	HNB	0.1992	1	0.2973	2
9	SAMP	-0.2271	5	-0.5695	8

Random Forest Company Ranking Based on Predicted Probability of Gain

- Mean Rank IC: 0.330968
- Top-3 Hit Rate: 0.375
- IC-IR: 1.053531

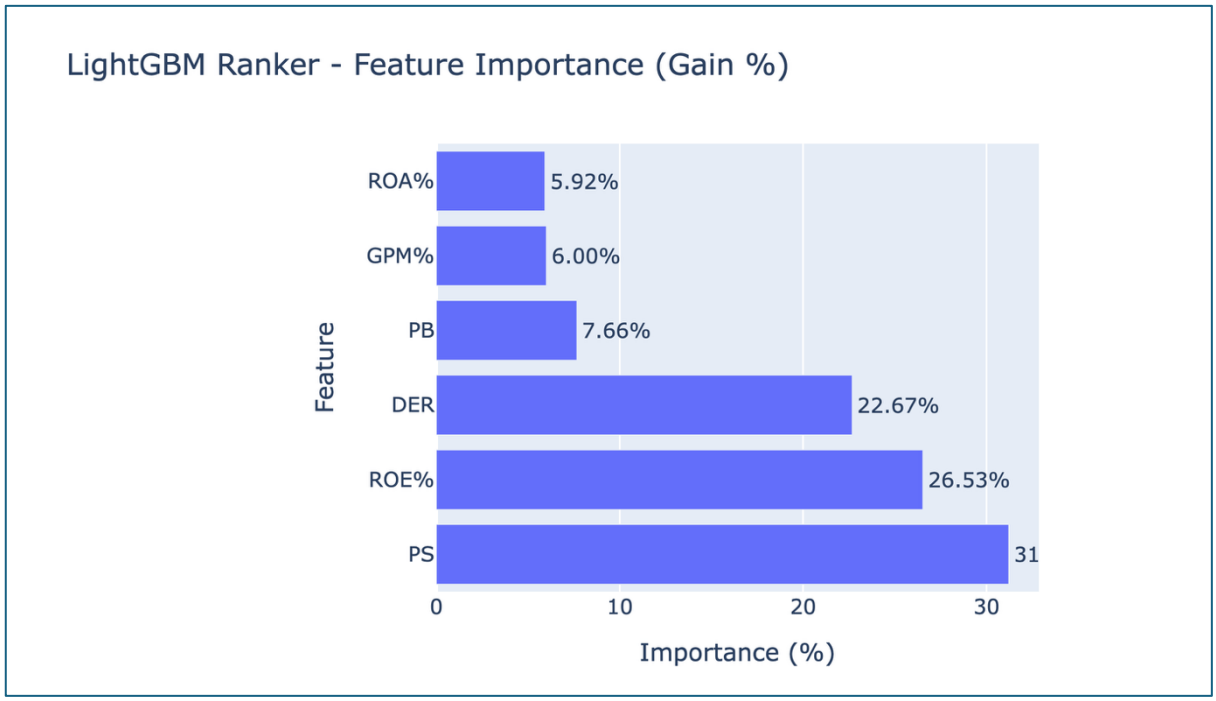
Company	Company_Name	RF_Prob	RF_Rank
0	CARG	0.9725	1
7	COMB	0.9575	2
9	SAMP	0.945	3
1	CTC	0.9375	4
8	HNB	0.9075	5
2	DIAL	0.89	6
4	MELS	0.8025	7
5	JKH	0.6025	8
3	DIST	0.5925	9
6	LOLC	0.4825	10

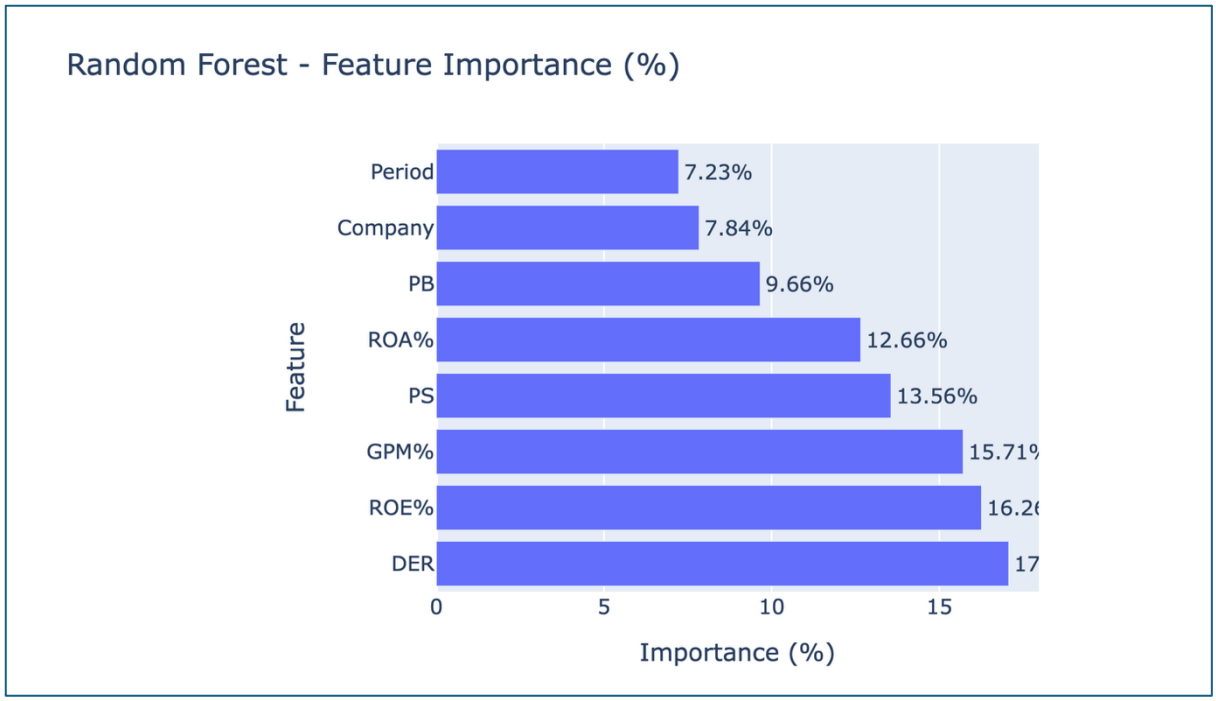


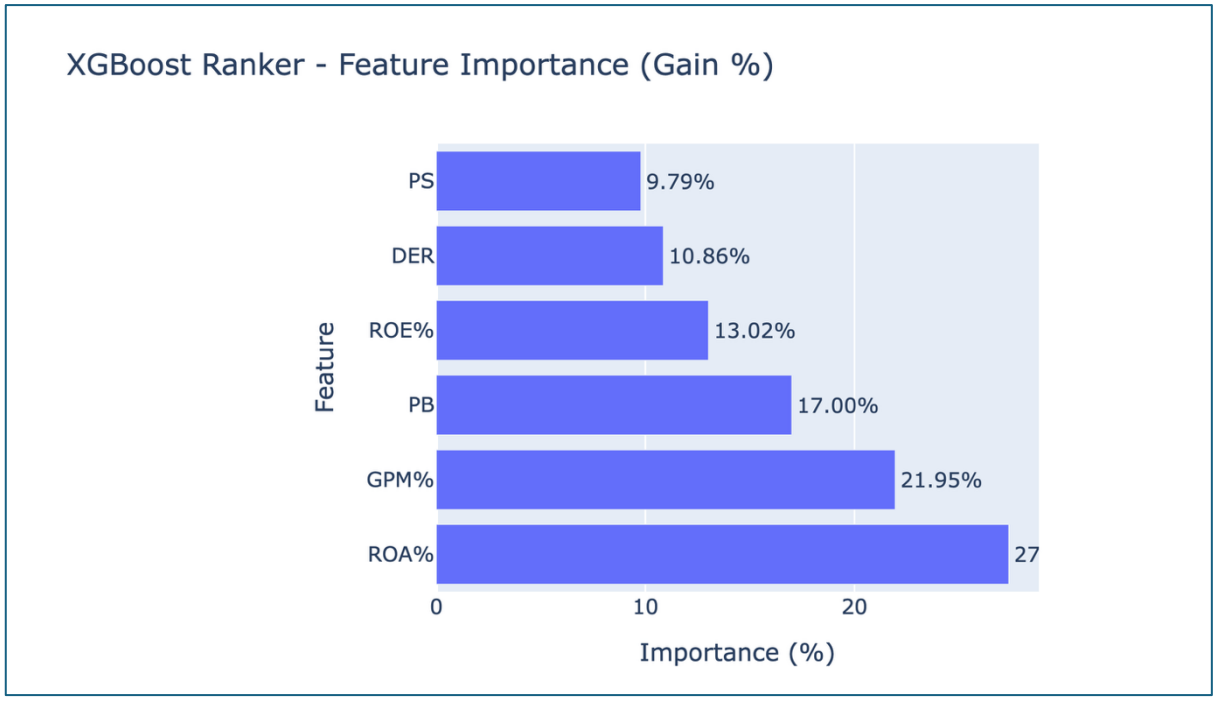
All Three Models

Company	Name	LightGBM Score	LightGBM Rank	XGBoost Score	XGBoost Rank	RF Score	RF Rank
8	HNB	0.1992	1	0.2973	2	0.9075	5
3	DIST	0.0070	2	0.1645	3	0.5925	9
2	DIAL	-0.0404	3	0.7255	1	0.8900	6
0	CARG	-0.0568	4	-0.1005	5	0.9725	1
9	SAMP	-0.2271	5	-0.5695	8	0.9450	3
7	COMB	-0.3206	6	-0.6443	9	0.9575	2
1	CTC	-0.4007	7	-0.0257	4	0.9375	4
5	JKH	-1.2669	8	-2.3218	10	0.6025	8
6	LOLC	-1.4557	9	-0.5496	7	0.4825	10
4	MELS	-1.5814	10	-0.2407	6	0.8025	7

Feature Importance




Random Forest - Feature Importance (%)
Period
Company
PB
ROA%
PS
GPM%
ROE%
DER
7.23%
7.84%
9.66%
12.66%
13.56%
15.71%
16.26
17
Feature
0
5
10
15
Importance (%)


XGBoost Ranker - Feature Importance (Gain %)
PS
DER
ROE%
PB
GPM%
ROA%
9.79%
10.86%
13.02%
17.00%
21.95%
27
Feature
0
10
20
Importance (%)

Combined Feature Importance Across Three Models

Feature	LightGBM Gain (%)	XGBoost Ranker Gain (%)	Random Forest (%)
ROE%	26.53%	13.02%	16.26%
PS	31.21%	9.79%	13.56%
DER	22.67%	10.86%	17.07%
ROA%	5.92%	27.38%	12.66%
GPM%	6.00%	21.95%	15.71%
PB	7.66%	17.00%	9.66%